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Paleomagnetism of the Moreton's Harbour Group, Northeastern Newfoundland Appalachians: Evidence for an Early Ordovician Island Arc Near the Laurentian Margin of Iapetus

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Paleomagnetic results have been obtained from mafic volcanic units in the upper part of the Moreton's Harbour Group, which is part of an accreted island arc terrane that is preserved in the Notre Dame Bay subzone of the Central Mobile Belt of the Newfoundland Appalachians. Detailed thermal and alternating field demagnetization reveals a stable characteristic component of magnetization, carried by magnetite, at a large number of sites in pillow basalts and coeval basaltic dikes. The intrusives contain both polarities, and a primary age for the characteristic magnetization is indicated by a positive contact test for one of the dikes, and by a positive structural test involving a correction for block rotations (strike-correction). The overall mean direction for the characteristic component after tectonic (tilt and strike) correction (flows and intrusives: $D=171^{o}$, $I=+22^{o}$, $k=22.6$, $\alpha 95=6.5^{o}$, pole 29°N, 135°E; flows: $D=166^{o}$, $I=+22^{o}$, $k=33.2$, $\alpha 95=6.5^{o}$) corresponds to an Early Ordovician paleolatitude for the arc of 11°S, which is indistinguishable from the expected paleolatitude of the North American margin. This implies that the arc formed at or near the margin of the craton. In contrast, the Avalon block, which formed the other margin of Iapetus, was widely separated from the arc and the craton at this time. The Early Ordovician paleolatitude of the arc terrane supports a tectonic model in which coeval ophiolitic sequences inboard of the arc were formed in a narrow ocean basin between the arc and the craton. Subsequent convergence between the arc and Laurentia in Middle Ordovician time resulted in closure of this narrow ocean basin and obduction of the back-arc basin oceanic crust onto the ancient margin of the craton, thus giving rise to the Taconic orogenic pulse in the Newfoundland Appalachians.

INTRODUCTION

The Appalachian Mountains of eastern North America preserve remnants of the Paleozoic Iapetus ocean that was located between Laurentia, comprising North America, Greenland plus northern Great Britain, and western Gondwana, consisting of South America and Africa [e.g., *Wilson*, 1966; *Dewey*, 1969; *Bird and Dewey*, 1970; *Harland and Gayer*, 1972; *McKerrow and Ziegler*, 1972; *Cocks and Fortey*, 1982; *Fortey and Cocks*, 1988; *Van der Voo*, 1988]. The Lower Paleozoic paleogeography of this ocean is complicated by the presence of more than one ocean basin [e.g., *Kidd*, 1977; *van der Pluijm and van Staal*, 1988; *Whalen*, 1989; *Bevier and Whalen*, 1990; *Stockmal et al.*, 1990].

The tectonic evolution of these basins and other tectonic elements of Iapetus, such as the continental margins, arcs and ocean islands, can be constrained by paleomagnetic data. However, widespread late Paleozoic remagnetization [*Creer*, 1968; *Van der Voo*, 1989; *Miller and Kent*, 1988; *McCabe and Elmore*, 1989] has often obscured primary directions in many critical units, which has limited the application of paleomagnetism to Appalachian paleogeography. Improved techniques, extensive sampling and careful analyses, however, have expanded the available database for eastern North America and in particular, the Newfoundland Appalachians.

The Newfoundland Appalachians are subdivided into the miogeocline (Humber zone), the Central Mobile Belt and the Avalon terrane (Figure 1, inset) [*Williams*, 1979; *Williams and Hatcher*, 1983; *van der Pluijm and van Staal*, 1988; *Williams et al.*, 1988]. The Humber zone recorded the formation and subsequent destruction of the northern margin of Iapetus [*Bradley*, 1989; *James et al.*, 1989]. A lower Paleozoic shelf facies is structurally overlain by Lower Ordovician ophiolite slices, such as the Humber Arm and Hare Bay allochthons, which were emplaced in Middle Ordovician time [*Stevens*, 1970; *Church and Stevens*, 1971; *Williams*, 1975]. Recent work has shown that thrusting in Middle Paleozoic time further modified this area [*Cawood and Williams*, 1988]. The Grenvillian basement that underlies the Humber zone outcrops in several places in western Newfoundland [*Williams*, 1978]. Off-shore, deep-seismic reflection data suggest that the Grenvillian basement extends a considerable distance under the neighboring Central Mobile Belt [*Keen et al.*, 1986; *Marillier et al.*, 1989]. The Central Mobile Belt (CMB; as defined by *van der Pluijm and van Staal*, 1988] preserves remnants of Iapetus, including Lower Paleozoic magmatic arcs, ocean floor sequences and a marine sedimentary deposits. *Williams et al.* [1988] subdivided this area into several (sub)zones. Structural as well as geophysical evidence indicates that the CMB is an allochthonous element [*Karlstrom et al.*, 1982; *Colman-Sadd and Swinden*, 1984; *van der Pluijm*, 1987; *Marillier et al.*, 1989; *Stockmal et al.*, 1990]. The boundary between the CMB and the Humber zone is marked by the Baie Verte-Brompton Line that is characterized by the occurrence of ophiolite assemblages [*Williams and St. Julien*, 1982]. The Avalon terrane, which represents the southern bordering continental block of Iapetus, was a stable tectonic element since the Late Proterozoic [*O'Brien et al.*, 1983; *Rast and Skehan*, 1983; *Krogh et al.*, 1988]. Its boundary with the CMB in northeastern Newfoundland is a major fault, the Dover Fault, which exhibits a complex movement history [*Kennedy et al.*, 1982]. Marine seismic reflection data across this boundary seem to indicate that the Dover fault penetrates into the mantle [*Keen et al.*, 1986; *Marillier et al.*, 1989].

Recent compilations of paleomagnetic data show intermediate

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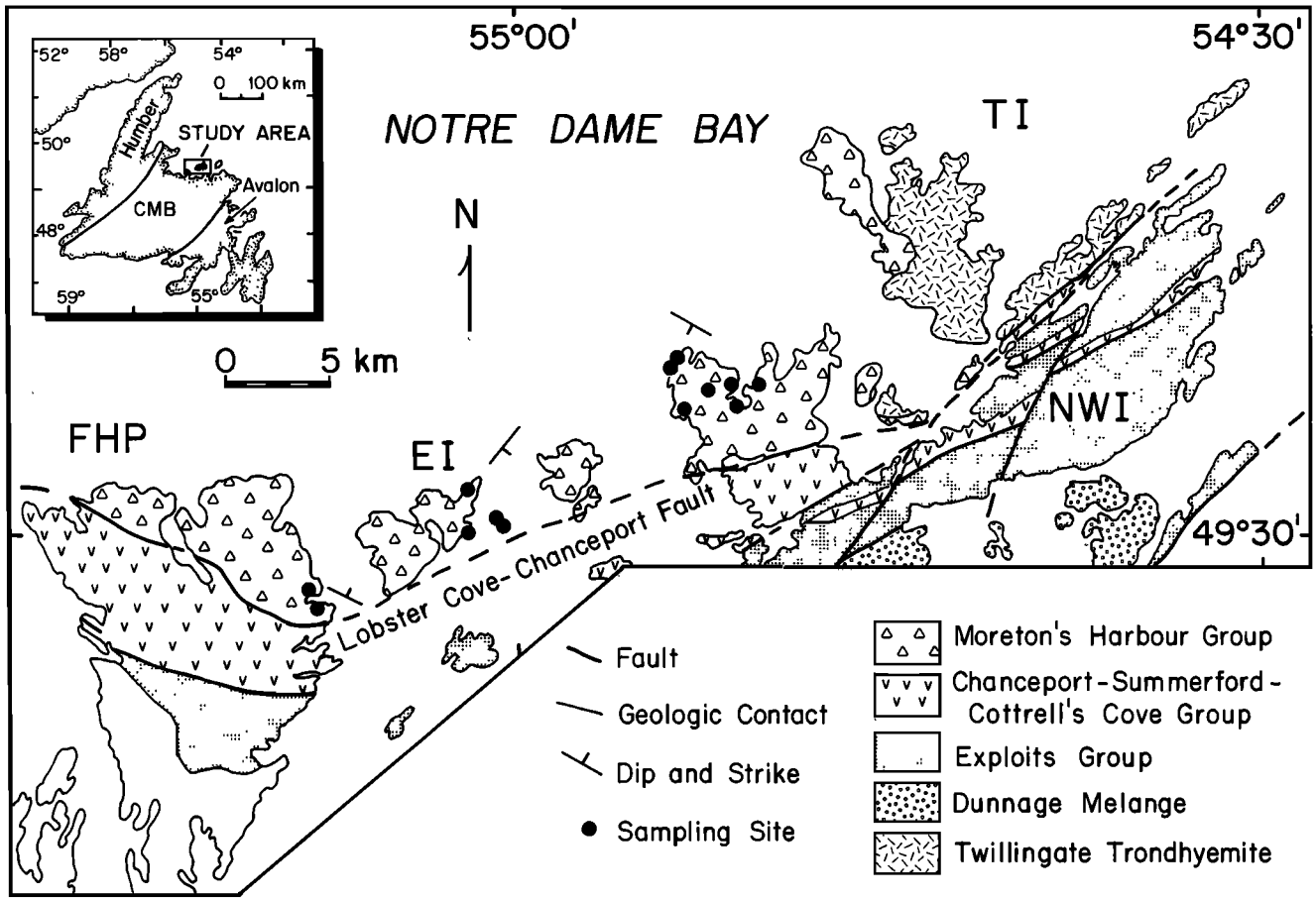


Fig. 1. Subdivision of the Newfoundland Appalachians (inset) and simplified geologic map of northeastern Newfoundland showing sampling sites. Abbreviations used are: EI, Exploits Island; FHP, Fortune Harbour Peninsula; NWI, New World Island; TI, Twillingate Island. Newfoundland subdivision after *van der Pluijm and van Staal* [1988] and regional geology after *Kean et al.* [1981] and *Williams et al.* [1988].

to high paleolatitudes (40-50°S) for the Avalon terrane in Ordovician times that are close to those of northern Gondwana [*Johnson and Van der Voo*, 1986; *Van der Voo*, 1988; *McCabe and Channell*, 1990; *van der Pluijm et al.*, 1990], implying that this continental block was located along the southern margin of Iapetus. This configuration is also supported by sedimentologic data [*Noblet and Lefort*, 1990]. During this time, the Appalachian margin of Laurentia was oriented approximately east-west and located at low paleolatitudes of 10-20°S [*Briden et al.*, 1988; *Van der Voo*, 1988]. Paleobiogeographic data for Laurentia and Avalon/Gondwana [*Cocks and Fortey*, 1982; *Neuman*, 1984; *Fortey and Cocks*, 1988] are in good agreement with such a wide separation between these continental blocks [*Van der Voo*, 1988].

In this paper we will present the results from a study of Lower Ordovician mafic volcanic units of the Moreton's Harbour Group of northcentral Newfoundland, which are part of an Upper Cambrian-Lower Ordovician magmatic arc sequence. These results place important constraints on the Lower Paleozoic paleogeography of Iapetus and tectonic evolution of the northern Appalachians. Preliminary paleomagnetic results for the Moreton's Harbour Group were reported previously by *Deutsch and Rao* [1977] in a reconnaissance study of selected Lower Paleozoic rock units in Newfoundland. However, their result was based on incomplete demagnetization of the samples (blanket alternating-field treatment only) and field tests for the age of magnetization were unavailable. In light of the subsequent evidence for widespread remagnetization of many rock units in the Appalachians mentioned earlier, the latter are especially important. Sampling for this study was extended over a much broader geographic area and stratigraphic range, which allows a fold test, baked-contact test and reversal test to be performed to constrain the age of the characteristic components of magnetization.

GEOLOGIC SETTING

The Upper Cambrian-Lower Ordovician Moreton's Harbour Group of northwestern Newfoundland is located in the Notre Dame Bay subzone [*Williams et al.*, 1988] of the Central Mobile Belt (Figure 1). *Dean* [1978] presented a summary description of the various units of the Moreton's Harbour Group that collectively form part of a Lower Paleozoic island arc sequence that was built onto oceanic crust [*Strong and Payne*, 1973; *Kean and Strong*, 1975; *Williams and Payne*, 1975]. Our sampling was concentrated in the upper units of the sequence, the Little Harbour Formation and the Western Head Formation. These units and fossiliferous correlatives to the west, the Cutwell and Western Arm Groups, are assigned a Lower Ordovician age [*Dean*, 1978]. Correlative units of the Moreton's Harbour Group farther to the southwest (e.g., Victoria Lake Group) indicate that formation of the Lower Ordovician arc locally

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occurred on continental crust, indicating a near continental margin setting for the arc [*Swinden and Thorpe*, 1984].

The Moreton's Harbour Group is separated from the Chanceport and Summerford Groups of New World Island and the Cottrell's Cove Group of Fortune Harbour Peninsula by the steeply-dipping Lobster Cove-Chanceport Fault (Figure 1) [*Dean and Strong*, 1977; *Williams et al.*, 1988]. In addition to marked lithologic, structural and geochemical contrasts across the Lobster Cove-Chanceport Fault [*Williams*, 1979; *Jacobi and Wasowski*, 1985; *Wasowski and Jacobi*, 1985; *Bostock*, 1988], faunal evidence indicates the presence of two distinct provinces [*Neuman*, 1984, 1988]. Units of the Moreton's Harbour Group have intermediate to steep southerly dips on northwestern New World Island and Exploits Island, and northerly dips on eastern Fortune Harbour Peninsula (Figure 1). Regional variation in strike direction and cleavage orientation suggest the possibility of block rotations postdating the main episode of folding and tilting, as will be discussed further below. Excellent preservation of pillow shapes enabled determination of the younging direction in flows.

SAMPLING

The paleomagnetic sampling sites are shown in Figure 1. The primary criteria for site selection were access and adequate control on the orientation of bedding (paleohorizontal). In addition, we tried to sample strata having different structural attitudes so that a paleomagnetic fold test could be performed. Based on such criteria, most of the sampling was concentrated in shoreline exposures of the pillow basalt sequences (16 sites) and the cross-cutting basaltic or felsic dikes (eight sites) within the Little Harbour and Western Head Formations. Paleohorizontal for the intrusive units was estimated from nearby host strata.

Obtaining accurate bedding orientations is often a problem in paleomagnetic studies of volcanic sequences. However, in the present study involving pillow basalts this problem was mitigated by the fact that individual pillows, in addition to indicating the younging direction, are typically flattened within the plane of bedding (intermediate/minimum diameter is about 1.5). This relationship was verified at a number of localities where interbedded tuffs were also present, particularly within the Little Harbour Formation. The consistency between stratification in the sediments and the average plane of pillow flattening at these localities gave us confidence in sampling other localities where only pillows are present. The plane of pillow flattening was determined by averaging measurements of the plane of elongation in many individual pillows, or by sighting along the estimated average elongation in large (three-dimensional) exposures of the flows. The errors associated with such measurements are assumed to be random, and we estimate the resultant uncertainties in bedding strike and dip angles to be $< 10^{o}$.

The intrusives were sampled together with flows at localities near the village of Moreton's Harbour and on Exploits Island. The intrusives were mainly dikes, but we also sampled a thin (2.5 m thick) basalt sill near Moreton's Harbour (site OLHd1'), which cuts a tuff horizon between pillow basalt units. Although the margins of the sill are generally parallel to the stratification in the sediments, the latter are locally contorted and partially incorporated by the sill, suggesting that the sill was intruded while the sediments were still soft. This evidence supports a similar age (Early Ordovician) for the sill and the host strata. The sill and the bounding sediments and pillow basalt units at this locality strike approximately WNW-ESE, and near the low-water line they abut against a N-S trending vertical dike (about 2.5 m wide) which was also sampled (site OLHd1). It is not clear from the discontinuous exposure whether the sill is connected to the dike, or is similarly cut by it. The pillow basalt units stratigraphically below and above the sill (sites OLH1 and OLH2, respectively) were sampled several meters (>5 m) from the N-S trending dike and along a strike-parallel profile approaching the dike margin, in part to perform a baked contact test for the age of magnetization in the dike. Ideally, if the pillow basalts near the dike were to show the same direction of magnetization as the dike, but different from the surrounding country rock, this would be evidence of a primary age of magnetization in the dike [e.g., *McElhinny*, 1973]. Even though the intrusives and volcanic strata are likely to be of similar age, such a test might still be useful if the host strata and intrusives have opposite-polarity magnetizations. The same sampling scheme was followed at a nearby locality where another N-S trending dike (site OLHd2) cuts pillow basalts of site OLH6. We sampled pillow basalts (OWH11) near a single basalt dike (site OWHd1) found on northeastern Exploit's Island; most of the volcanic samples were located 8-10 m from the dike, but two flow samples were collected immediately adjacent to the dike margin for the purposes of comparison.

All of the samples were collected with a portable gas-powered core drill, and were oriented with a Brunton compass and inclinometer. As verified in the field and the laboratory, the low magnetization intensities of the volcanics did not significantly affect the compass measurements. A minimum of six pillows were sampled at each of the flow sites, usually over about 2 m of section, and in most cases two samples were collected from each pillow. Samples were collected from the chilled margins as well as from the coarser grained interiors of the dikes.

LABORATORY METHODS

Standard paleomagnetic specimens (2.2 x 2.5 cm diameter) were prepared from the oriented core samples, and their remanent magnetization was measured with an ScT two-axis cryogenic magnetometer or a Schonstedt SSM-1A spinner magnetometer. Specimens were demagnetized by stepwise alternating field (AF) and/or thermal methods, using a Schonstedt GSD-1 single-axis AF demagnetizer and a Schonstedt TSD-1 furnace, and results for each specimen were plotted on orthogonal vector diagrams [*Zijderveld*, 1967] as well as on equal-area projections. The principal component method of *Kirschvink* [1980] was used to calculate best-fit "demagnetization lines" and "planes" (great-circles) for linear and planar demagnetization trajectories identified from the graphs. Great-circle intersections were determined for components with overlapping spectra, and individual sample directions and great-circles were combined to calculate the means at several sites using the method of *Bailey and Halls* [1984].

The magnetic carriers in the Moreton's Harbour volcanics were investigated using a number of different rock magnetic techniques, as well as scanning electron microscopy (SEM). Isothermal remanent magnetization (IRM) acquisition experiments were performed on unheated core specimens, at room temperature (25°C), in applied fields of up to 1.4 Tesla (T). Hysteresis properties and Curie temperatures (saturation magnetization (J_s) versus temperature) of the samples were measured on a vibrating sample magnetometer (VSM) at the University of Minnesota. The hysteresis measurements were made on core specimens at room temperature, while J_s versus temperature measurements were done with crushed rock samples (approximately 50 mg), or magnetic mineral separates,

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in an atmosphere of helium gas. Polished sections of the volcanics were examined with the SEM and the (Fe/Ti) composition of individual opaque grains was estimated using energy dispersive analysis (EDA).

PALEOMAGNETIC RESULTS

The Moreton's Harbour Group flows and intrusives have similar natural remanent magnetization (NRM) intensities, ranging from 10^{-4} to 10 A/m. The NRM directions for each rock type are generally well-grouped at the site level; however, these directions, plotted in Figure 2, show considerable between-site scatter in both geographic and bedding coordinates. The demagnetization results for the flows and intrusives, and the analysis of the characteristic directions in both rock types, are presented in the separate subsections that follow.

Flows

Typical thermal and AF demagnetization results for the flows are shown in Figures 3-5. The thermally demagnetized samples of Figures 3*a*-3*c* reveal that a low-temperature component of magnetization is removed first by about 500°C. The directions for first removed components in the volcanics are generally grouped about the present day field (PDF) direction for central Newfoundland in geographic coordinates, indicating that these are most likely viscous magnetizations of recent origin (Figure 4). However, the low-temperature components in a few samples

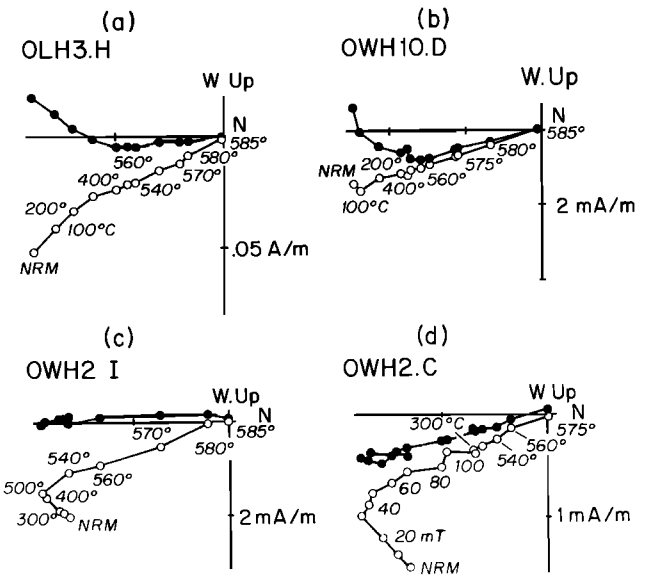


Fig. 3. Orthogonal vector diagrams [*Zijderveld*, 1967] showing (*a-c*) stepwise thermal demagnetization and (*d*) combined alternating field (af) and thermal demagnetization of flow samples from the Moreton's Harbour Group, plotted after tilt and strike correction. Open (solid) symbols represent projections of the vector endpoints on the north-south vertical (horizontal) plane. The treatment steps are labeled in degrees celsius (C), or milliTesla (mT). The remanent magnetization intensity is indicated along the axes in amperes per meter (A/m) or milliamperes per meter (mA/m).

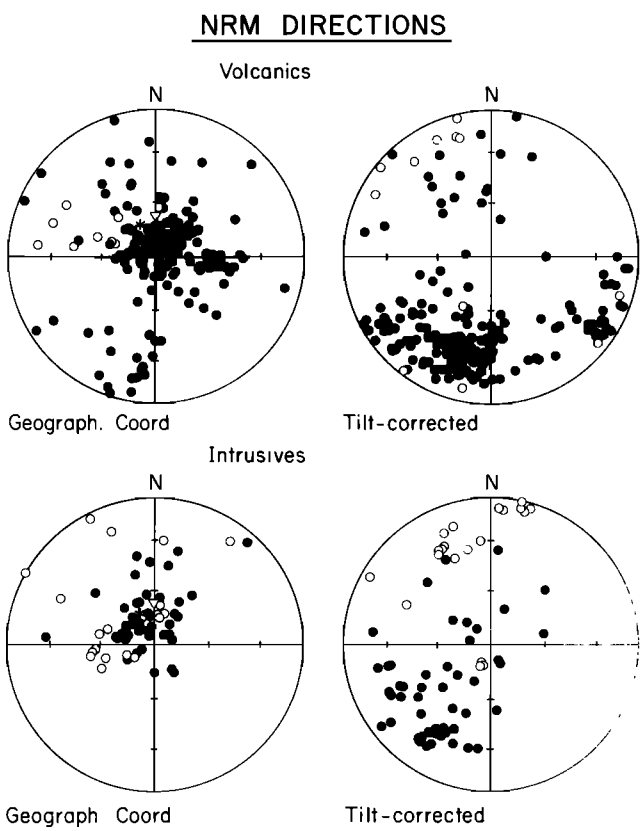


Fig. 2. Equal-area projections of the natural remanent magnetization (NRM) directions for the flow samples (volcanics) and intrusive samples, shown in geographic coordinates and after tilt correction. Solid (open) circles denote lower (upper) hemisphere projections; the asterisks in the projections on the left indicate the present-day geomagnetic field (PDF) direction in the study area, while the inverted triangle represents the present dipole field direction for this location.

from sites OLH6 and OWH11 yielded westerly and up, and north(west)erly and shallower directions, respectively. As we discuss below, the components removed first at these sites may be partial thermal or thermochemical magnetizations dating from the time of emplacements of nearby dikes within the Moreton's Harbour Group (Early Ordovician). Following removal of the low-temperature component, the results for samples depicted in Figure 3 show a characteristic component of magnetization having a southerly and shallow direction, which decays linearly to the origin by 590°C. The maximum blocking temperature for the characteristic magnetization in these samples is close to the Curie temperature of pure magnetite (585°C), suggesting that the characteristic remanence is carried by low-titanium (low-Ti) magnetite.

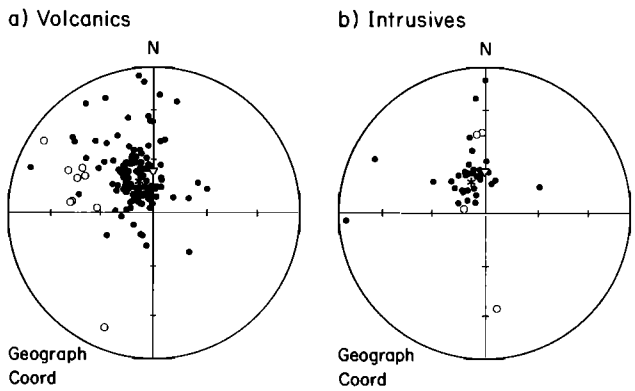


Fig. 4. Equal-area projections of the sample directions for low temperature components in (*a*) the Moreton's Harbour flows and (*b*) intrusives. The same plotting conventions are used as in Figure 2.

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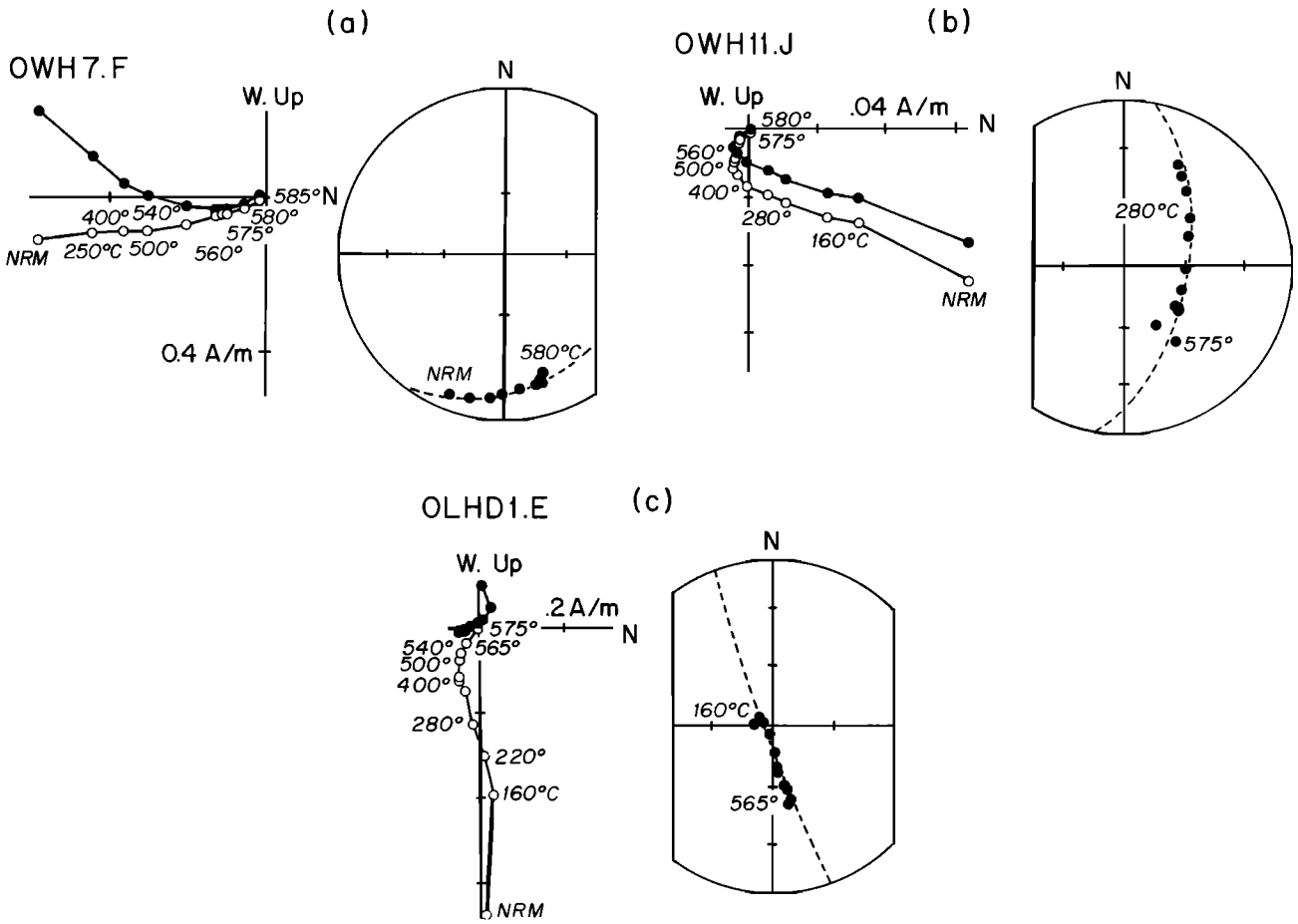


Fig. 5. Orthogonal vector diagrams and equal-area projections showing (*a*, *b*) thermal demagnetizations of flow and (*c*) intrusive samples which yielded great-circles, plotted after tilt and strike correction. The best fit plane (great-circle) to the demagnetization trajectory, and the beginning and end points of the demagnetization interval used, are identified in the equal area projections. The same plotting conventions are used as in the previous diagrams.

The combined alternating field (AF)/thermal demagnetization of one flow sample from site OWH2 is shown in Figure 3*d*. The initial AF treatment of this sample, which was done as a pilot experiment, removes a soft PDF component by 35 mT and appears to isolate the southerly and shallow-downward characteristic component of magnetization (e.g., compare Figure 3*b*); however, maximum AF fields of 100 mT do not completely remove the remanence. Subsequent thermal treatment of this sample results in linear decay of the remaining magnetization to the origin by 590°C, clearly revealing the characteristic magnetization and its direction. Although the carrier of the characteristic remanence in OWH2.C exhibits a high stability to AF fields, the maximum blocking temperature of 590°C again suggests that the carrier is magnetite. Because AF treatment of pilot samples yielded incomplete demagnetizations, this treatment method was discontinued and the remaining flow samples were all thermally treated to ensure accurate identification of the remanence components and their carriers.

The demagnetization diagrams for some of the flow samples (e.g., Figure 5) show typical low-temperature linear trends below 500°C. Above this temperature the demagnetization trajectories are curved, indicating the simultaneous removal of more than one component having overlapping blocking temperatures. The presence of the characteristic component in these samples is suggested by the fact that the remanence direction in each sample migrates along a great circle path toward a shallow southerly direction during progressive demagnetization, even though stable endpoint directions for this component are clearly not reached. The great-circle trajectories of Figure 5 confirm that just two components of magnetization are present in each of these samples, including the apparent low-temperature component initially isolated on the vector plots below 500°C; the blocking temperatures for this component evidently extend up to 590°C resulting in the observed overlap with the characteristic magnetization. Following *Halls* [1976, 1978] the best-fit plane to the demagnetization trajectory (great-circle) for each sample (e.g., Figure 5) is inferred to contain both components of magnetization. Although the characteristic component could not be isolated in such samples, the great circle for each sample was combined with similar data (directions and/or planes) for other samples to estimate the mean direction of the characteristic component at the site level (cf., Table 1) [*Bailey and Halls*, 1984]. It is noted that site OWH11 is the only site at which the characteristic component could not be isolated in any of the samples (Table 1). However, owing to a large dispersion in the directions for "low-temperature" components at this site, the best-fit great circles for individual samples converge toward a southerly and shallow direction, and their resultant best-fit intersection is taken as the direction of the characterisic component for site OWH11 (e.g., Table 1).

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TABLE 1. Site Mean Paleomagnetic Data for the Characteristic Component of the Moreton's Harbour Group, Northeastern Newfoundland

Site	Bedding S/D	$n_d/n_g/n_o$	I-S D/I, deg	T-C D/I, deg	T-C&S-C D/I, deg	k	α_{95}
Flows							
OLH1	105/67S	13/00/13	095/+65	168/+25	167/+25	50.5	5.9
OLH2	105/67S	13/00/17	078/+63	166/+33	165/+33	44.9	6.3
OLH3	105/67S	09/00/09	080/+69	173/+30	172/+30	53.5	7.1
OLH4	105/67S	11/00/11	131/+67	174/+12	173/+12	70.9	5.5
OLH5	105/67S	11/00/11	117/+70	174/+18	173/+18	29.1	8.6
OLH6	100/67S	09/00/10	119/+79	179/+19	183/+19	29.3	9.7
OWH1	313/15N	09/00/10	187/+16	183/+28	154/+28	150.0	4.2
OWH2	298/28N	08/00/08	193/+03	191/+30	177/+30	31.2	9.4
OWH3	023/70S	09/00/10	046/+72	096/+12	177/+12	61.6	6.6
OWH4	063/55S	08/00/08	096/+47	118/+07	159/+07	125.2	5.0
OWH6	063/55S	10/00/10	098/+53	123/+10	164/+10	78.2	5.5
OWH7	140/70S	02/07/11	138/+58	196/+18	160/+18	66.9	10.1
OWH8	140/70S	05/06/11	116/+43	179/+31	143/+31	154.2	4.6
OWH9	118/62S	05/03/09	134/+70	187/+21	173/+21	45.6	9.7
OWH10	128/80S	05/02/07	122/+60	187/+12	163/+12	53.9	9.3
OWH11	220/07N	00/11/11	095/+40	091/+46	155/+46	116.1	9.1
Mean Flows (16/16)			122/+62 k=7.6 α_{95}=14.4	164/+25 k=6.6 α_{95}=15.5	166/+22 k=33.2 α_{95}=6.5		
Intrusives							
OLHd1	105/67S	04/03/09	85/+70	174/+28	173/+28	55.3	9.9
OLHd1'	105/67S	06/00/08	013/-74	014/-07	013/-07	238.7	4.3
OLHd2	100/67S	09/00/09	262/-60	337/-29	341/-29	88.3	5.5
OLHd3	105/67S	04/03/07	100/+67	170/+29	169/+29	42.8	11.2
OLHd4	140/65S	04/02/07	156/+72	212/+19	176/+19	34.5	13.4
OLHd5	140/65S	03/02/06	304/+79	241/+22	205/+22	44.4	14.1
OLHd6	105/67S	06/00/12	207/+65	200/-02	199/-02	40.9	10.6
OWHd1*	220/07N	06/00/08	-	-	-	1.2	90.0
Mean Flows and Intrusives			124/+68 k=8.6 α_{95}=11.0	173/+25 k=6.3 α_{95}=13.1	171/+22 k=22.6 α_{95}=6.5		

Bedding (paleohorizontal) is indicated by strike, dip (S/D) and direction of dip (N or S); n_d and n_g specify the number of sample directions and great circles used to calculate the site mean, respectively, versus the total number of samples that were demagnetized for each site (n_o). The site mean directions are reported in geographic coordinates (I-S), after tilt correction (T-C), and after tilt and strike correction (T-C&S-C; see text); D/I are declination and inclination of the mean directions. The polarity directions for sites OLHd1' and 2 were reversed to calculate the overall mean directions; k and α_{95} are statistical parameters associated with the means.

The direction from site OWHd1 was not used in the mean calculation because of its low precision.

Intrusives

Samples of intrusive rocks from the majority of sites likewise appear to contain ancient components of magnetization. Thermal demagnetization of samples from two of the basalt dikes (e.g., Figures 6*a* and 6*b*) reveal very similar behavior to that found for the flows: following initial removal of a low temperature component of magnetization below 500°C, a southerly and shallow downward characteristic component isolated in these samples decays linearly to the origin by 590°C. In a few samples from site OLHd1 the magnetization trends toward the expected direction for the characteristic component during thermal demagnetization, but this component could not be fully isolated. As before, best-fit demagnetization planes (great circles) were calculated for such samples, and these were combined with similar data from the same site to obtain the mean for the characteristic component. The magnetization in these dikes is inferred to be carried by magnetite.

In contrast to the previous two examples, the demagnetization of samples from a third basaltic dike (Figure 6*c*), and from the one sill that was sampled (Figures 6*d* and 6*e*), yield univectorial magnetizations having northerly and shallow upward directions. The magnetizations in these samples are interpreted to represent opposite-polarity characteristic components. As is evident from the thermal demagnetizations shown in Figures 6*c* and 6*d*, the blocking temperatures for the magnetization in these samples are approximately uniformly distributed between the NRM and

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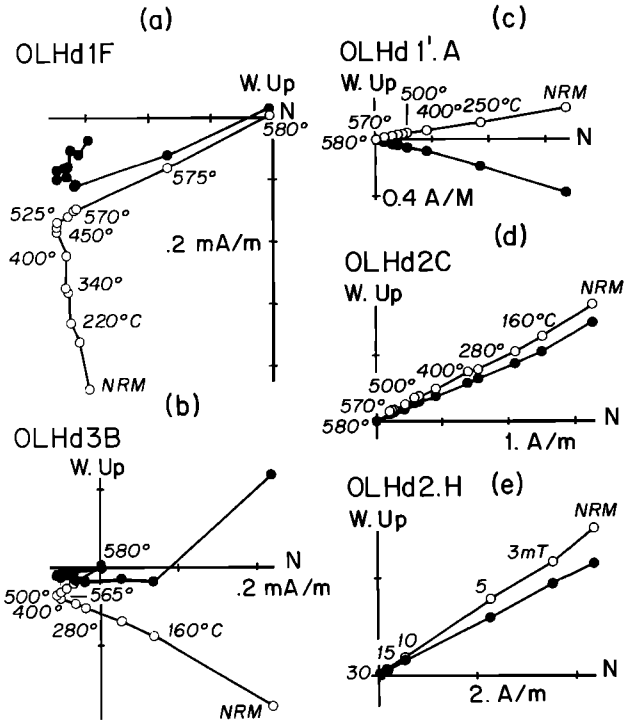


Fig. 6. Orthogonal vector diagrams showing (*a-d*) thermal and (*e*) alternating field demagnetizations of intrusive samples. The same plotting conventions are used as in previous diagrams.

590°C. The AF demagnetization of Figure 6*e* shows that the same component of magnetization in a second sample from the sill is removed in relatively low fields of 20 mT. The low AF stability exhibited in Figure 6*e*, in particular, suggests that the univectoral magnetizations in these samples are probably carried by multidomain magnetite.

The directions for high-temperature components in samples from one additional dike, located on Exploit's Island, were found to vary from northerly and shallow upward, to steep (up or down), to southerly and shallow downward. As expected, the mean for this site (OWHd1) has a very low precision and large circle of confidence (see Table 1). Such a large scatter of directions could conceivably be explained by slow cooling of the host dike in a time-varying paleomagnetic field (e.g., transitional), or it might indicate the effect of a lighting strike.

As noted above, the pillow basalts adjacent to the dikes at a couple of different localties were sampled to assess the age of magnetization in the dikes with baked contact tests. The most conclusive results were obtained for the dike sampled at Site OLHd2, which has a northerly and shallow-upward characteristic direction of magnetization (Figure 7). The mean direction found for the dike is plotted in Figure 7*b* along with the directions observed during demagnetization for three pillow basalt samples obtained at different distances from the dike margin; the vector demagnetization diagrams for the same three flow samples are shown in Figures 7*c*-7*e*. As seen from the figure, the flow sample collected closest to the dike margin (OLH6.B; about 5 cm) contains a univectorial magnetization having the same northerly and shallow-up direction as the dike. This behavior contrasts with the multicomponent magnetizations seen in the other samples which were collected farther from the dike margin. In the middle sample (OLH6.D; Figure 7*d*), located 0.5 m from the dike, a component approximately parallel to the dike direction is removed first by 550°C revealing an additional southerly and shallow characteristic component. Similarly, following removal of a shallow westerly low-temperature component, a well-defined southerly and shallow characteristic direction is isolated between 560°C and 580°C in sample OLH6.H (Figure 7*e*). The low-temperature component in this latter sample lies along a great-circle approximately midway between the northerly direction in the dike and the southerly characteristic direction in the flows and is considered to be a composite direction resulting from the simultaneous removal of these two components. The magnetizations in all three flow samples of Figure 7 are completely removed by 580°C, indicating that the observed magnetizations are probably carried by magnetite.

The shallow northerly directions in the flow samples in Figure 7 are interpreted to be secondary overprint magnetizations acquired when the dike of site OLHd2 was intruded, most likely resulting from localized heating of the surrounding country rock. The univectoral magnetization in sample OLH6.B is clearly consistent with a thermal origin, since the temperature at the time of intrusion near the dike probably exceeded 580°C, and thus was capable of completely resetting any magnetization carried by magnetite in this sample. Similarly, removal of the overprint magnetizations in samples OLH6.D and H by 550°C appears to indicate partial resetting of their magnetizations, consistent with expected lower peak temperatures farther from the dike margin where these samples were obtained. Because overprint magnetizations (observed or inferred) were found in all of the samples from site OLH6 (up to 4 m from the dike margin), completely unheated samples were not available for direct comparison with the heated ones along this particular profile. Nevertheless, the exclusive occurrence of such overprints in the vicinity of this dike and nowhere else in the study area demonstrates a positive contact test, indicating that the magnetization in the dike itself is primary. Based on this evidence, the magnetization of this dike is therefore considered to be Early Ordovician in age. Moreover, following *McElhinny* [1973], we note that this positive contact test also provides indirect evidence for the age of the stable southerly and shallow-downward characteristic magnetization in the surrounding pillow basalt country rock. Thus the inferred partial to complete resetting of the characteristic magnetization in the immediate vicinity of this dike implies that the characteristic component was present when the dike was emplaced. In this particular case, since the dike is only slightly younger than the country rock, this evidence suggests that the characteristic magnetization in the flows has a primary, or at least Early Ordovician age, as well.

At a second locality near the village of Moreton's Harbour, southerly and shallow characteristic directions were found in the dike (site OLHd1) and the surrounding country rock (sites OLH1, 2) regardless of distance from the intrusive margin. While it is clear that both rock types acquired their magnetizations in similar paleomagnetic fields and therefore probably at the same time, the contact test in this case is inconclusive. Nevertheless, because the two rock types probably have similar ages, this certainly allows the possibility that the characteristic components in each are primary. If true, this would imply that the dike emplacement in this region was diachronous and, in particular, that it spanned at least one paleomagnetic polarity transition.

The third contact test tried on Exploit's Island is precluded because the sample directions in the dike at this locality (site OWHd1) are widely scattered. While the cause of this directional scatter is uncertain, we infer that the unusually large dispersion in the low-temperature directions from adjacent flows (site OWH11) probably have a similar origin. It is noted that the possible causes for the scatter suggested above (e.g.,

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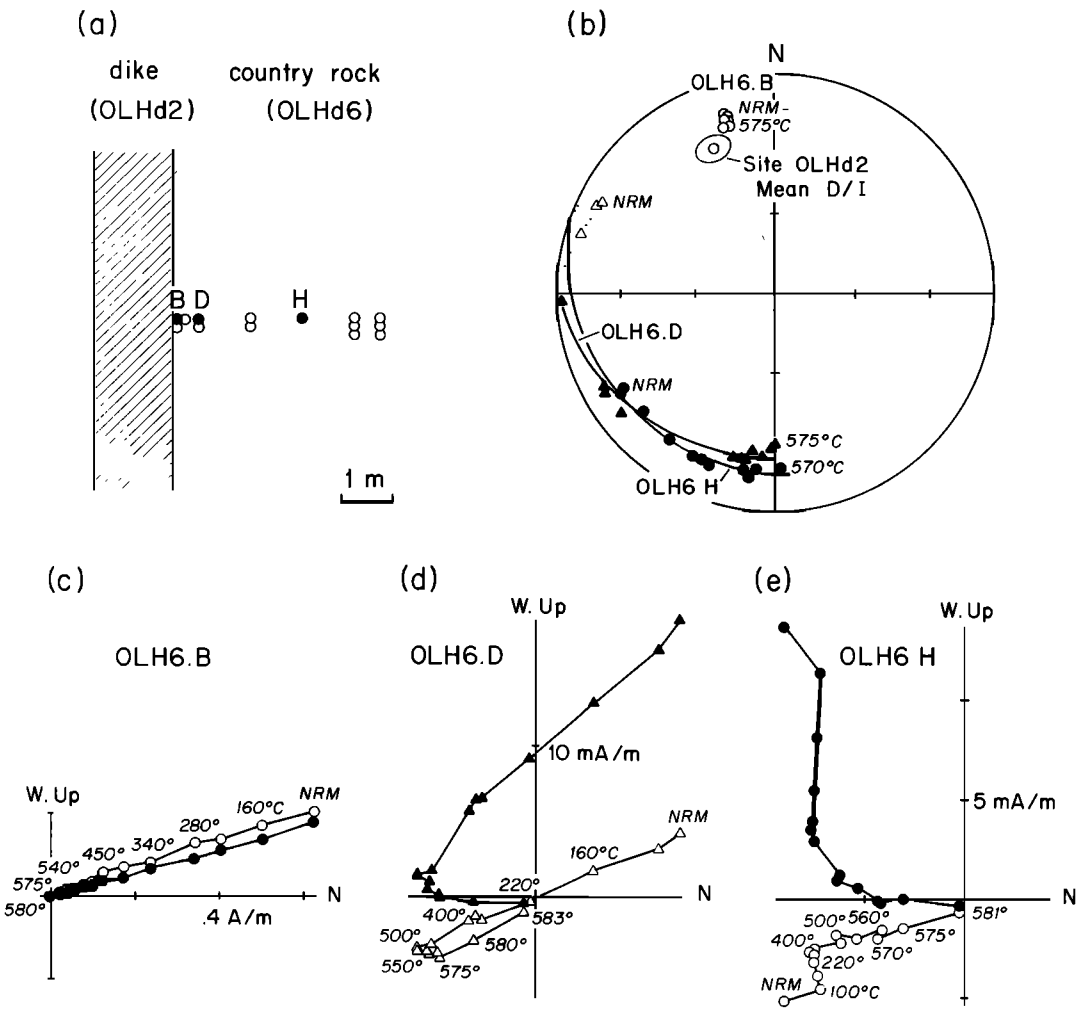


Fig. 7. Summary of the baked contact test results for the dike sampled at site OLHd2. The dike and the locations of samples taken in the adjacent country rock (site OLH6) are shown schematically in (*a*); the locations of representative country rock samples discussed in detail in the text (OLH6B, D, and H; labeled in figure) are denoted by the solid circles. The thermal demagnetization results for these three country rock samples are compared with the site mean characteristic direction in the dike in the equal-area projection of (*b*), and are illustrated in the orthogonal vector diagrams (*c-e*). The best fit planes (great circles) to the demagnetization trajectories for samples OLH6D and H are shown in the equal area projection; for clarity, triangle symbols were used to identify the demagnetization trajectory for sample OLH6D. All plots are shown after correction for tectonic tilt.

lightning strike, cooling/resetting in a transitional field) would very likely have affected both rock units at this locality.

Fold Test for the Characteristic Magnetization

The equal area plots in Figure 8 illustrate the effects of structural corrections on the site mean directions for the characteristic components in the Moreton's Harbour Group volcanics. The directions for both flows and intrusives are shown, and they are grouped together for the purposes of statistical analysis because these rock units and their magnetizations appear to be of similar ages. Except for the mean from site OWHd1, which is not considered reliable because of its large uncertainty, all of the site mean directions listed in Table 1 are included in Figure 8.

As seen in Figure 8*a*, the site means are scattered in geographic coordinates. A number of volcanic and dike sites obtained from steeply dipping strata yield directions in geographic coordinates having easterly declinations and intermediate to steep inclinations; in contrast, the nearly flatlying volcanics from Fortune Harbour Peninsula have well-defined southerly and very shallow site means. The steeply upward directions in Figure 8*a* were obtained from the intrusives of sites OLHd1' and OLHd2, which yielded the positive contact test. Upon simple tilt correction (Figure 8*b*) the directions from sites on New World Island and Fortune Harbour Peninsula, which have northwest-southeast trending strikes, converge towards a southerly and shallow-downward direction or a northerly and shallow-upward (opposite-polarity) direction, suggesting a prefolding age for the characteristic magnetization at these sites. Yet, the intermediate localities on Exploits and Hare Islands, which have northeasterly strikes (Figure 1), yield shallow easterly directions, thus resulting in a large dispersion in the overall distribution of tilt-corrected site means. As expected, the fold test based on the site means in Figures 8*a* and 8*b* is not significant ($k1/k2 = 1.12$).

While the tilt-corrected site means for the two groups of sites represented in Figure 8*b* reveal considerable scatter in declination, their respective inclinations are very similar. This

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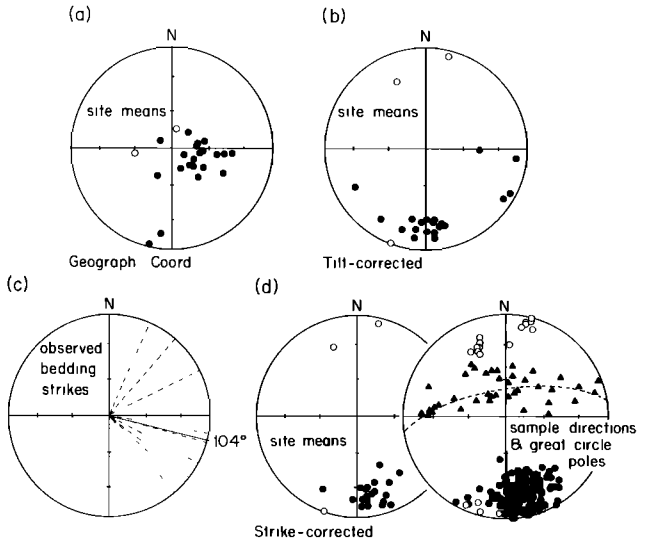


Fig. 8. Equal-area projections showing the characteristic directions for the Moreton's Harbour Group volcanics. The site mean directions for the flows and intrusives are plotted (*a*) together in geographic coordinates and (*b*) after tilt correction. The observed bedding strikes for individual sampling sites (dotted lines) are shown in (*c*), along with the mean reference strike used for the strike correction (104°; solid line); the number of bedding strikes shown is less than the number of sites because several of the sites have the same structural attitude (cf. Table 1). The site means and sample data are shown after strike correction in (*d*): the triangles in this diagram denote the poles to individual sample great-circles, while the circle symbols indicate sample directions or site means; the great-circle shown for reference in the figure was drawn perpendicular to the overall mean direction and is not an attempt to fit the great-circle data.

evidence leads us to suspect that the two portions of the study area have undergone a vertical-axis rotation relative to one another, in addition to the observed tilting. Although declination differences such as those of Figure 8*b*, could arise through failure to recognize other modes of deformation (e.g., tilting about an inclined axis), true vertical-axis rotations are the only way to explain the large declination differences particularly between the directions for the nearly flatlying (i.e., gently dipping) sites within each group [e.g., MacDonald, 1980]. Comparison of the bedding strikes in the vicinity of Exploits and Hare Islands (Figure 1) shows that they are deflected in the same sense and by the same amount, relative to the other sites, as the paleomagnetic declinations. This relationship is consistent with the interpretation that vertical-axis rotations occurred after the tilting of the volcanic strata. Moreover, the variation in bedding strikes thus provides an estimate of the relative rotation between the different sites and this information can be used to correct the sites and their paleomagnetic directions to their predeformation (prerotation) relative configuration (i.e., "strike correction"). We especially wanted to examine whether such a correction significantly improved the grouping of site mean directions for the characteristic component, as would be expected if this magnetization predates the deformation. As presented below, this application of the strike correction is equivalent to performing a fold test about a vertical axis.

The bedding strikes for those sites that yielded the characteristic directions are plotted in the equal-area diagrams of Figure 8*c*. As noted previously, these display considerable variation, ranging from northeasterly to southeasterly trends. In applying a strike correction the standard procedure is to correct all of the sites is to a common reference direction, which is therefore assumed to represent their prerotation relative configuration. For the purpose of the statistical test, the choice of the latter is arbitrary; as in the present case, where the true bedding trends prior to rotation are unknown, the reference direction is usually taken to be the mean (average) of the observed strikes (104°; see Figure 8*c*). The strike correction applied to a given site thus corresponds to the angular difference between its strike and the reference direction. It is important to emphasize that the strike corrections calculated in this manner reflect rotations, but probably do not represent the actual rotation amounts of the sites. Such rotation amounts, as well as true overall mean direction prior to deformation, could only be determined if the pre-rotational bedding trends were known.

Figure 8d shows the grouping of site means and sample directions, as well as their much improved Fisherian distributions, resulting from application of this strike correction. The increase in the precision parameter (k) of the overall mean after strike correction (e.g., Table 1), relative to geographic or tilt-corrected coordinates, indicates that the improved grouping of the site means evident in the figures is statistically significant at the 99% confidence level [*McElhinny*, 1973]. This result provides further conclusive evidence that the characteristic magnetization in the Moreton's Harbour Group was acquired prior to deformation, with an overall mean direction of $D=166^{\circ}$, $I=+22^{\circ}$ ($k=33.2$, $\alpha_{95}=6.5^{\circ}$) for flows only and $D=171^{\circ}$, $I=+22^{\circ}$ ($k=22.6$, $\alpha_{95}=6.5^{\circ}$) for flows and intrusives combined.

MAGNETIC MINERALOGY

The results of the IRM experiments and J_s vs. T measurements for representative volcanic samples are shown in Figure 9. These samples all yielded very similar IRM acquisition results, in which saturation magnetization is reached in applied fields of less than 0.2 T, thus indicating the presence of only magnetite. The J_s vs. T curves show that each of the samples has a single Curie temperature in the range of 570-580°C. Thus the magnetites have similar near end-member (low-Ti) compositions in all of the volcanic samples. The hysteresis measurements for these and other volcanic samples were found to yield consistently low remanence coercivities ($H_{cr}<0.1$ T), although the saturation magnetization intensities differed by some three orders of magnitude. For samples with high ferromagnetic contents (e.g., OWH8.D, OLHd2.H), the hysteresis ratios indicated grain sizes near the pseudo-single domain/multidomain boundary ($H_{cr}/H_c = 4.0$, $J_{rs}/J_s = 0.1$); whereas the samples with lower ferromagnetic contents (e.g., OWH6.F) had ratios indicating grain sizes in the finer pseudo-single domain range ($H_{cr}/H_c = 2.0$, $J_{rs}/J_s = 0.3$) [*Day* et al., 1977].

Individual opaque grains were readily observable using the SEM only in samples having high ferromagnetic contents. Consistent with the rock magnetic measurements, Fe-oxides in such samples were abundant and comprised roughly equant grains having sizes ranging from 1 to 100 μm. The different size grains in the flows appear to be homogeneous and, as inferred from energy dispersive analysis (EDA; 15 keV) they have fairly uniform compositions with Fe:Ti ratios of about 4:1, which are in approximate agreement with the oxide compositions of modern-day oceanic basalts [e.g., *Johnson*, 1979). Although well-defined exsolution textures were not seen in the opaque grains from any of the flow samples, the high Curie temperatures indicate that they must be exsolved into Ti-

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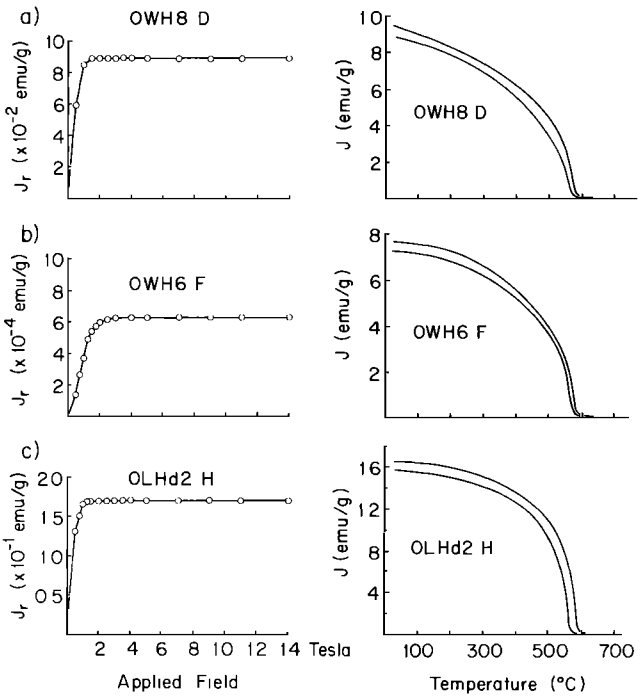


Fig. 9. Isothermal remanent magnetization (IRM) acquisition results, and thermomagnetic (J_s versus T) curves for (*a*) a flow sample with high-ferromagnetic content, (*b*) a flow sample with low-ferromagnetic content, and (*c*) a representative dike sample. J_r is the remanent magnetization measured at room temperature. All three samples saturate rapidly in applied fields of less than 0.3 Tesla, and have Curie temperatures near 580°C, indicating the presence of only (low-titanium) magnetite. All of the results were obtained using whole rock samples except the thermomagnetic curve for OWH6.F, which was made using a crushed magnetic separate.

rich and remanence carrying Ti-poor phases on a submicroscopic scale. It is noted that ultra-fine exsolution of this type has been suggested by *Smith and Banerjee* [1986] to explain similar textural and magnetic characteristics of modern ocean ridge basalts. In contrast to the flows, relict textures indicative of high-temperature exsolution were commonly observed in opaque grains of the intrusive samples. The Ti-rich portions (lamellae) of these grains appear to be replaced by sphene, but the Ti-poor portions are intact.

The textural evidence in the intrusive samples clearly indicates a primary (deuteric) origin for the magnetic carriers in these rocks. However, the evidence in the case of the flows is more equivocal. Thus, the low-Ti magnetic carriers could have formed as a result of deuteric exsolution under relatively low-oxygen activities, or the exsolution may have occurred post-depositionally due to hydrothermal alteration of the flows. Nevertheless, as required by the field tests (e.g., baked-contact test), any postdepositional alteration of the flows must have occurred very soon after deposition in order to predate the intrusion of the associated dikes and sills of the Moreton's Harbour Group.

TECTONIC IMPLICATIONS

The mean direction for the characteristic magnetization in the Moreton's Harbour Group after tilt and strike correction is $D/I=171^{\circ}/+22^{\circ}$ ($k=22.6$, $\alpha_{95}=6.5^{\circ}$; Table 1), with a corresponding paleopole at 29°N, 135°E (Table 2). This direction and paleopole include sites in flows as well as in intrusives. Excluding the sites in intrusives, which were tilt and strike corrected using the bedding orientation in the host rock, results in a direction with a slightly different declination, but indistinguishable inclination ($D/I=166^{\circ}/+22^{\circ}$, $k=33.2$, $\alpha_{95}=6.5^{\circ}$; Table 1). Because we used a mean strike to apply the strike correction, the declination only reflects this reference

TABLE 2. Early Ordovician Paleomagnetic Pole Positions for the Moreton's Harbour Group and Iapetus-Bordering Continental Elements

Rock unit and Location	Age	South Pole	k	α_{95}	Q	Ref
North America						
Everton Formation., N. Arkansas	Ol/m	07S, 356E	-	-	5	1
St. George Group, W. Newfoundland	Ol	18S, 332E	144	9	5	2
Oneota Dolomite, IA, MN, WI	Ol	10S, 346E	18	12	6	3
Mean		12N, 165E	39	20		
Gondwana						
Taylor Valley dikes, Antarctica	470 Ma	36N, 015E	75	11	4	4
Sor Rondane Intrusives, Antarctica	480 Ma	12N, 014E	-	5	3	5
Jinduckin Formation, N. Australia	Ol	31N, 001E	10	11	6	6
Graafwater Formation, S. Africa	Ol	28N, 014E	25	9	6	7
Ntonya Ring Structure, Malawi	464 Ma	28N, 345E	1054	2	5	8
Hook Intrusives, Zambia	500 Ma	14N, 336E	13	36	3	9
Mean		26N, 001E	20	15		
Moreton's Harbour Group, Newfoundland	Ol	29N, 135E dp, dm = 3.6, 6.9	23	7	6	10

O, Ordovician; l, lower; m, middle; k and α_{95} are statistical parameters associated with the means; Q, Quality factor [see *Van der Voo*, 1989]. References: 1, *Farr and Sprowl* [1990]; 2, *Deutsch and Prasad* [1987]; 3, *Jackson and Van der Voo* [1985]; 4, *Manzoni and Nanni* [1977]; 5, *Zijderveld* [1968]; 6, *Luck* [1972]; 7, *Bachtadse et al.* [1987]; 8, *Briden* [1968]; 9, *Brock* [1967]; 10, this paper.

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strike and is not a uniquely determined parameter. Thus, the declination direction will not be used in the tectonic interpretation of our results. Note that while a strike correction strictly rotates the beds about a vertical axis, the mean inclination is slightly different after (22°) versus before (25°) this correction. The shallower mean inclination after strike correction is attributed to the improved clustering of the site mean directions after strike correction (see Figure 8 and corresponding k and α_{95}). The overall mean inclination of 22° indicates an Early Ordovician paleolatitude for the study area of 11°, with an associated uncertainty given by the dp of the paleopole (dp=3.6°, Table 2).

The overall mean direction found in this study (e.g., Table 1) is very similar to the one reported for the Moreton's Harbour Group by Deutsch and Rao [1977] (D/I=176°/+16°, k=9, α_{95}=16.5°). Because these authors sampled only a single locality, they were unable to detect block rotations. The similar mean declination found by those authors is fortuitous, and only indicates that the particular locality they sampled has not been rotated significantly relative to our reference strike for the area. Our result for the Moreton's Harbour Group confirms the shallow paleomagnetic inclination and demonstrates the primary age of the magnetization, which together dictate a low paleolatitudinal position during Early Ordovician time.

The positions of Laurentia, Avalon/Armorica and Gondwana in Early Ordovician times have been determined based on paleomagnetic results, and these are shown in Figure 10. The data used for this reconstruction are listed in Table 2. Also shown in Figure 10 is the Lower Ordovician position of the magmatic arc based on the results in this paper. We represent the ancient position of this paleogeographic element by a band around the calculated paleolatitude to emphasize the absence of paleolongitudinal constraints. The overlap between the Appalachian margin of Laurentia and the possible positions of the magmatic arc shows that the latter was formed at or near the northern margin of Iapetus. We note that, because of the large uncertainty in the mean pole of North America (α_{95}=20°; Table 2), the area of overlap may be considerably less. Moreover, the presence of the Lower Paleozoic carbonate platform sequence in western Newfoundland indicates that the arc was somewhat distant from the continental margin. Thus, we conclude that active island-arc volcanism occurred near the Laurentian margin in Early Ordovician time. The paleogeography in Figure 10 also illustrates that the main portion of Iapetus in Ordovician times was located to the south (ancient coordinates) of the magmatic arc. As a consequence, collision of the arc with Laurentia, which traditionally has been associated with the Taconic Orogeny [e.g., *Williams*, 1979; *Bradley*, 1989], does not represent the main closure of Iapetus [cf. *Williams and Hatcher*, 1983; *Casey et al.*, 1985]. Rather, we favor an interpretation in which the Taconic pulse recorded the closure of a small, short-lived ocean basin (Iapetus I) that was formed between the volcanic arc and Laurentia [*van der Pluijm et al.*, 1990]. Collision of the magmatic arc with Laurentia resulted in closure of Iapetus I and the emplacement of ophiolites on the Laurentian margin. Today, portions of ophiolites are preserved in, for example, the Humber Arm Allochthon of western Newfoundland [*Williams*, 1979]. The E-W orientation of the Laurentian margin and the absence of longitudinal constraints from paleomagnetic data prevent the recognition of possible strike-slip movement along this margin [e.g., *Keppie*, 1989]. Subsequent Appalachian orogenic events are associated with continued closure of Iapetus (Iapetus II) and the collision of Avalon/Armorica with Laurentia ('Caledonian' and Acadian pulses), and the final collision between Laurentia and Gondwana (Alleghenian pulse).

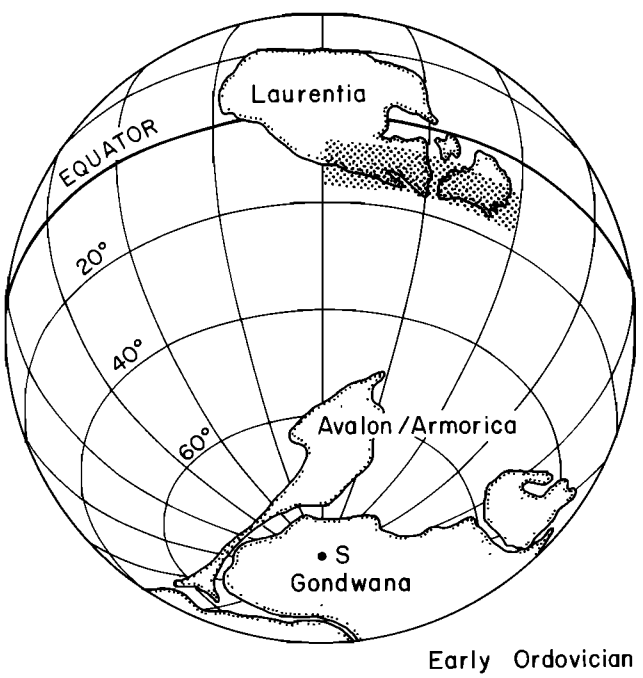


Fig. 10. Early Ordovician paleogeographic reconstruction of Laurentia, Avalon/Armorica, and Gondwana. The position of these continental elements is based on paleomagnetic data from the references listed in Table 2. The possible location of the magmatic arc, which is represented by pillow basalts of the upper Moreton's Harbour Group, is indicated by a band around the mean paleolatitude (11°S; Table 2). See text for discussion.

CONCLUSIONS

Pillow basalts of the Lower Ordovician Little Harbour and Western Head Formations of the Moreton's Harbour Group, which are located in the Notre Dame Bay subzone of the Central Mobile Belt of the northcentral Newfoundland Appalachians, preserve a stable, high-blocking temperature characteristic component of magnetization that is carried by magnetite. The characteristic direction in flows and coeval intrusives (D=171°, I=+22°, k=22.6, α_{95}=6.5°) passes fold, reversal and baked-contact tests, which support a primary age for the magnetization; the corresponding paleomagnetic pole is at 29°N, 135°E. Based on the criteria listed in *Van der Voo* [1989] we assign quality factor Q=6 to this result.

The mean direction for the Moreton's Harbour Group corresponds to a paleolatitude of ca. 11°S. This paleolatitude is indistinguishable form the position of the Appalachian margin of Laurentia in Early Ordovician times, which implies that the main part of Iapetus was located between the magmatic arc and Avalon/Armorica/Gondwana. We conclude that arc activity, which was associated with closure of Iapetus, occurred near the northern margin of this ocean basin. Collision of this arc with Laurentia and ophiolite obduction were recorded by the Taconic orogenic pulse.

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